LEVEL 20 · PART IX

Cost Optimization

20

Goal: Reduce cloud spend without sacrificing reliability by treating cost as an engineering metric.

RECOMMENDED TIME
2-3 weeks

TOPICS
9

PRACTICE / CAPSTONE
Audit a sample AWS/Kubernetes environment: tag costs, find waste, rightsize, add lifecycle policies and propose a savings plan with numbers.

Topics

#	Topic	What you will learn
1	FinOps **p.3705**	inform/optimize/operate, tagging, showback/chargeback, unit economics, Cost Explorer
2	Reserved Instances and Savings Plans **p.3717**	commitments, coverage vs utilization, break-even
3	Spot Instances **p.3729**	interruption handling, diversification, Spot on K8s/ASG, suitable workloads
4	Autoscaling for Cost **p.3743**	scale-in, scheduled scaling, scale-to-zero, off-hours shutdown
5	Rightsizing **p.3755**	utilization analysis, Compute Optimizer, VPA recommendations, risks
6	Storage Lifecycle Policies **p.3767**	S3 tiers, Intelligent-Tiering, snapshot cleanup, log retention
7	Network Egress Costs **p.3779**	inter-AZ/region/internet transfer, NAT cost, VPC endpoints, CDN offload
8	Database Costs **p.3791**	instance sizing, storage/IOPS, replicas, Aurora Serverless, caching to reduce load
9	Kubernetes Infrastructure Costs **p.3803**	bin packing, requests inflation, Kubecost/OpenCost, node pools, Karpenter
★	Level capstone project **p.3819**	Audit a sample AWS/Kubernetes environment: tag costs, find waste, rightsize, add lifecycle policies and propose a savings plan with numbers.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

